



Emoji Movie Review



Team

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Problem Description

People increasingly use emojis to express their feelings about movies and TV shows online. In this project, we will explore emoji usage patterns and correlations between specific emojis and movie ratings.

We will investigate whether a movie’s rating can be predicted based solely on a given set of emojis. Additionally, we’ll explore the meanings of emojis by comparing them to the accompanying text and examine relationships between emojis that frequently appear together.

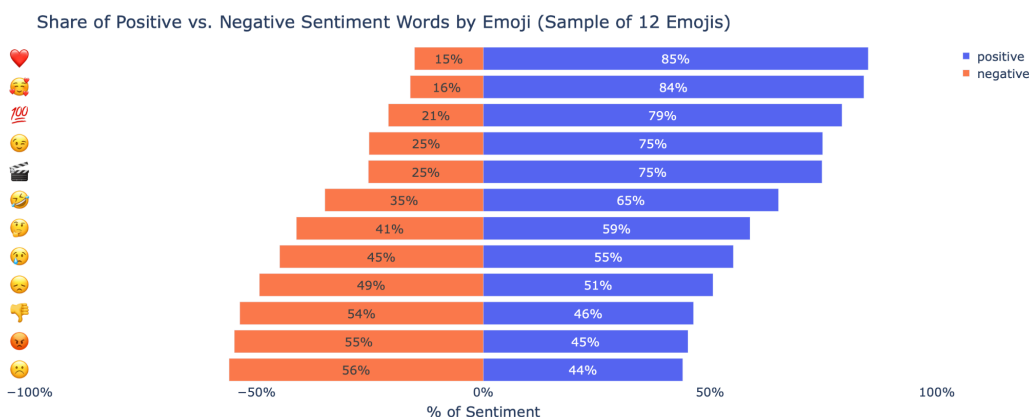
Data

We found a dataset (on [HuggingFace](#)) that combines posts about movies scraped from Reddit, with data from that movie’s listing on Amazon.

- **Size:** the dataset contains 5.3M reviews (1.65GB). Out of them, 34K reviews contain emojis.
- **Fields:** for every review, the dataset includes
 - The post’s title and text content
 - The discussed movie name, as appears in Amazon
 - The movie’s score on Amazon
 - Other structured features of the movie, including its duration, genre, cast, etc.

Visualizations

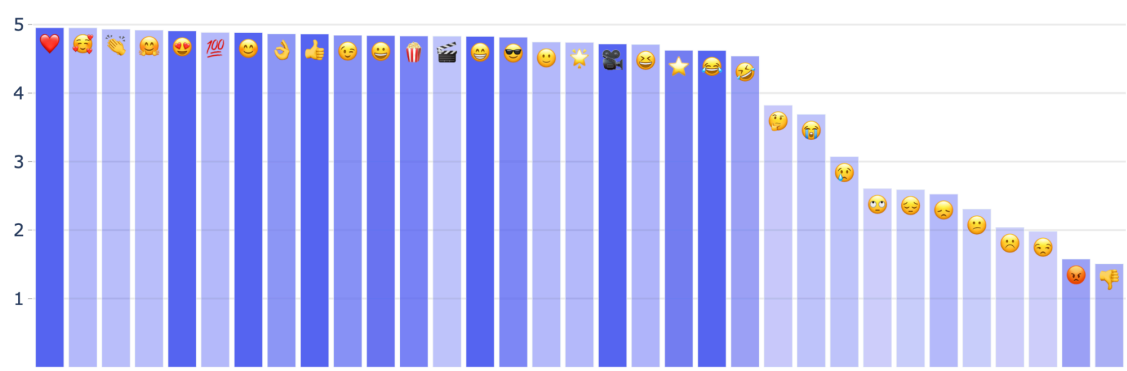
First, we expect some emojis to generally be used in a more positive sentiment, and others in a more negative one. We analyzed the sentiment of reviews using predefined lists of positive and negative words (taken from [Kaggle](#)). For each emoji, we looked at the number of positive and negative words it appeared in conjunction with.



Many emojis followed the pattern we expected, for example, “positive” emojis like “❤️” appeared more with words deemed positive.

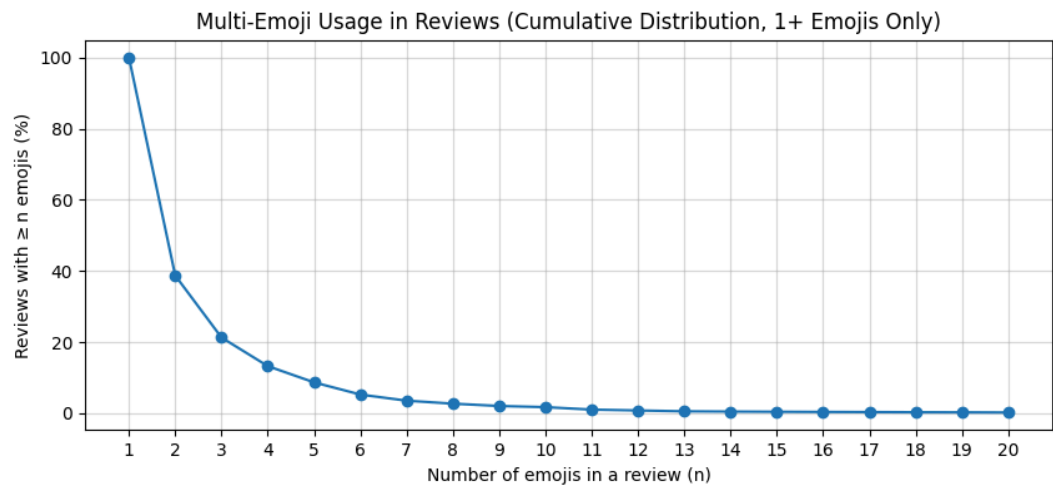
Second, we also expect that movies with more “positive” emojis in their reviews would generally be higher rated. We grouped reviews by the emojis that appeared in them and averaged movie ratings within each group (*again, only a subset of the emojis is shown*).

Average Ratings by Emoji Used in Reviews of That Movie (Opacity Indicates Number of Reviews)



As expected, emojis like “❤️” were linked to higher-rated movies, while emojis like “😡” were linked to lower ratings.

Lastly, while playing around with the data, we noticed that many reviews with emojis actually include more than one. For each n , we computed the percentage of reviews (out of those with at least one emoji) that had more than n emojis.



As shown, about 40% of reviews have 2 or more emojis, and around 20% have 3 or more. We could use this later to analyze the relationships between emojis and build a “community graph” of emojis.